

Animal Classification System

```
import java.util.ArrayList;

// Base Class
class Animal {
    String name;
    String species;

    // Constructor
    Animal(String name, String species) {
        this.name = name;
        this.species = species;
    }

    // Method
    void displayDetails() {
        System.out.println("Name: " + name);
        System.out.println("Species: " + species);
    }
}

// Derived Class - Mammal
class Mammal extends Animal {
    boolean hasFur;

    // Constructor
    Mammal(String name, String species, boolean hasFur) {
        super(name, species);
        this.hasFur = hasFur;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Mammal Details ---");
        System.out.println("Name: " + name);
        System.out.println("Species: " + species);
        System.out.println("Has Fur: " + hasFur);
    }
}

// Derived Class - Bird
class Bird extends Animal {
    double wingSpan;

    // Constructor
    Bird(String name, String species, double wingSpan) {
        super(name, species);
        this.wingSpan = wingSpan;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Bird Details ---");
        System.out.println("Name: " + name);
        System.out.println("Species: " + species);
        System.out.println("Wing Span: " + wingSpan + " meters");
    }
}

// Main Class
public class AnimalClassificationSystem {
    public static void main(String[] args) {

        // Collection of Animal references
        ArrayList<Animal> animals = new ArrayList<>();

        // Creating objects
        Mammal m1 = new Mammal("Tiger", "Mammal", true);
        Bird b1 = new Bird("Eagle", "Bird", 2.5);

        // Adding objects to collection
        animals.add(m1);
        animals.add(b1);

        // Demonstrating Polymorphism
        for (Animal a : animals) {
            a.displayDetails();
        }
    }
}
```

Banking System

```
import java.util.ArrayList;

// Base Class
class BankAccount {
    String accountNumber;
    String accountHolderName;
    double balance;

    // Constructor
    BankAccount(String accountNumber, String accountHolderName, double balance) {
        this.accountNumber = accountNumber;
        this.accountHolderName = accountHolderName;
        this.balance = balance;
    }

    // Method
    void displayDetails() {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Account Holder Name: " + accountHolderName);
        System.out.println("Balance: " + balance);
    }
}

// Derived Class - SavingsAccount
class SavingsAccount extends BankAccount {
    double interestRate;

    // Constructor
    SavingsAccount(String accountNumber, String accountHolderName, double balance, double interestRate) {
        super(accountNumber, accountHolderName, balance);
        this.interestRate = interestRate;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Savings Account Details ---");
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Account Holder Name: " + accountHolderName);
        System.out.println("Balance: " + balance);
        System.out.println("Interest Rate: " + interestRate + "%");
    }
}

// Derived Class - CurrentAccount
class CurrentAccount extends BankAccount {
    double overdraftLimit;

    // Constructor
    CurrentAccount(String accountNumber, String accountHolderName, double balance, double overdraftLimit) {
        super(accountNumber, accountHolderName, balance);
        this.overdraftLimit = overdraftLimit;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Current Account Details ---");
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Account Holder Name: " + accountHolderName);
        System.out.println("Balance: " + balance);
        System.out.println("Overdraft Limit: " + overdraftLimit);
    }
}

// Main Class
public class BankingSystem {
    public static void main(String[] args) {

        // Collection of BankAccount references
        ArrayList<BankAccount> accounts = new ArrayList<>();

        // Creating objects
        SavingsAccount s1 =
            new SavingsAccount("SB101", "Aniket", 50000, 5.5);

        CurrentAccount c1 =
            new CurrentAccount("CA201", "Rahul", 75000, 20000);

        // Adding objects to collection
        accounts.add(s1);
        accounts.add(c1);

        // Demonstrating Polymorphism
        for (BankAccount acc : accounts) {
            acc.displayDetails();
        }
    }
}
```

Education Course System

```
import java.util.ArrayList;

// Base Class
class Course {
    String courseName;
    int courseId;
    int duration;

    // Constructor
    Course(String courseName, int courseId, int duration) {
        this.courseName = courseName;
        this.courseId = courseId;
        this.duration = duration;
    }

    // Method
    void displayDetails() {
        System.out.println("Course Name: " + courseName);
        System.out.println("Course ID: " + courseId);
        System.out.println("Duration: " + duration + " months");
    }
}

// Derived Class - OnlineCourse
class OnlineCourse extends Course {
    String platform;

    // Constructor
    OnlineCourse(String courseName, int courseId, int duration, String platform) {
        super(courseName, courseId, duration);
        this.platform = platform;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Online Course Details ---");
        System.out.println("Course Name: " + courseName);
        System.out.println("Course ID: " + courseId);
        System.out.println("Duration: " + duration + " months");
        System.out.println("Platform: " + platform);
    }
}

// Derived Class - OfflineCourse
class OfflineCourse extends Course {
    String location;

    // Constructor
    OfflineCourse(String courseName, int courseId, int duration, String location) {
        super(courseName, courseId, duration);
        this.location = location;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Offline Course Details ---");
        System.out.println("Course Name: " + courseName);
        System.out.println("Course ID: " + courseId);
        System.out.println("Duration: " + duration + " months");
        System.out.println("Location: " + location);
    }
}

// Main Class
public class EducationCourseSystem {
    public static void main(String[] args) {

        // Collection of Course references
        ArrayList<Course> courses = new ArrayList<>();

        // Creating objects
        OnlineCourse o1 = new OnlineCourse(
            "Java Programming",
            101,
            6,
            "Udemy"
        );

        OfflineCourse o2 = new OfflineCourse(
            "Data Structures",
            202,
            4,
            "MITAOE Pune"
        );

        // Adding objects to collection
        courses.add(o1);
        courses.add(o2);

        // Demonstrating Polymorphism
        for (Course c : courses) {
            c.displayDetails();
        }
    }
}
```

```

Employee Management System

import java.util.ArrayList;

// Base Class
class Employee {
    String name;
    int employeeId;
    double salary;

    // Constructor
    Employee(String name, int employeeId, double salary) {
        this.name = name;
        this.employeeId = employeeId;
        this.salary = salary;
    }

    // Method
    void displayDetails() {
        System.out.println("Employee Name: " + name);
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Salary: " + salary);
    }
}

// Derived Class - Manager
class Manager extends Employee {
    int teamSize;

    // Constructor
    Manager(String name, int employeeId, double salary, int teamSize) {
        super(name, employeeId, salary);
        this.teamSize = teamSize;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Manager Details ---");
        System.out.println("Name: " + name);
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Salary: " + salary);
        System.out.println("Team Size: " + teamSize);
    }
}

// Derived Class - Developer
class Developer extends Employee {
    String programmingLanguage;

    // Constructor
    Developer(String name, int employeeId, double salary, String programmingLanguage) {
        super(name, employeeId, salary);
        this.programmingLanguage = programmingLanguage;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Developer Details ---");
        System.out.println("Name: " + name);
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Salary: " + salary);
        System.out.println("Programming Language: " + programmingLanguage);
    }
}

// Main Class
public class EmployeeManagementSystem {
    public static void main(String[] args) {

        // Collection of Employee references
        ArrayList<Employee> employees = new ArrayList<>();

        // Creating objects
        Manager m1 = new Manager("Aniket", 101, 75000, 10);
        Developer d1 = new Developer("Rahul", 102, 60000, "Java");

        // Adding objects to collection
        employees.add(m1);
        employees.add(d1);

        // Demonstrating Polymorphism
        for (Employee emp : employees) {
            emp.displayDetails();
        }
    }
}

```

```

Food Ordering System

import java.util.ArrayList;

// Base Class
class FoodItem {
    String itemName;
    double price;
    String category;

    // Constructor
    FoodItem(String itemName, double price, String category) {
        this.itemName = itemName;
        this.price = price;
        this.category = category;
    }

    // Method
    void displayDetails() {
        System.out.println("Item Name: " + itemName);
        System.out.println("Price: " + price);
        System.out.println("Category: " + category);
    }
}

// Derived Class - FastFood
class FastFood extends FoodItem {
    int preparationTime;

    // Constructor
    FastFood(String itemName, double price, String category, int preparationTime) {
        super(itemName, price, category);
        this.preparationTime = preparationTime;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Fast Food Details ---");
        System.out.println("Item Name: " + itemName);
        System.out.println("Price: " + price);
        System.out.println("Category: " + category);
        System.out.println("Preparation Time: " + preparationTime + " minutes");
    }
}

// Derived Class - Beverage
class Beverage extends FoodItem {
    String servingSize;

    // Constructor
    Beverage(String itemName, double price, String category, String servingSize) {
        super(itemName, price, category);
        this.servingSize = servingSize;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Beverage Details ---");
        System.out.println("Item Name: " + itemName);
        System.out.println("Price: " + price);
        System.out.println("Category: " + category);
        System.out.println("Serving Size: " + servingSize);
    }
}

// Main Class
public class FoodOrderingSystem {
    public static void main(String[] args) {

        // Collection of FoodItem references
        ArrayList<FoodItem> foodItems = new ArrayList<>();

        // Creating objects
        FastFood f1 = new FastFood(
            "Burger",
            120,
            "Fast Food",
            15
        );

        Beverage b1 = new Beverage(
            "Cold Coffee",
            90,
            "Drink",
            "Large"
        );

        // Adding objects to collection
        foodItems.add(f1);
        foodItems.add(b1);

        // Demonstrating Polymorphism
        for (FoodItem item : foodItems) {
            item.displayDetails();
        }
    }
}

```

```

Gaming Character System

import java.util.ArrayList;

// Base Class
class Character {
    String name;
    int health;
    int attackPower;

    // Constructor
    Character(String name, int health, int attackPower) {
        this.name = name;
        this.health = health;
        this.attackPower = attackPower;
    }

    // Method
    void attack() {
        System.out.println(name + " attacks with power " + attackPower);
    }
}

// Derived Class - Warrior
class Warrior extends Character {
    int armorStrength;

    // Constructor
    Warrior(String name, int health, int attackPower, int armorStrength) {
        super(name, health, attackPower);
        this.armorStrength = armorStrength;
    }

    // Overriding Method
    @Override
    void attack() {
        System.out.println("\n--- Warrior Attack ---");
        System.out.println(name + " attacks with Sword!");
        System.out.println("Attack Power: " + attackPower);
        System.out.println("Armor Strength: " + armorStrength);
    }
}

// Derived Class - Mage
class Mage extends Character {
    int magicPower;

    // Constructor
    Mage(String name, int health, int attackPower, int magicPower) {
        super(name, health, attackPower);
        this.magicPower = magicPower;
    }

    // Overriding Method
    @Override
    void attack() {
        System.out.println("\n--- Mage Attack ---");
        System.out.println(name + " casts a Magic Spell!");
        System.out.println("Attack Power: " + attackPower);
        System.out.println("Magic Power: " + magicPower);
    }
}

// Main Class
public class GamingCharacterSystem {
    public static void main(String[] args) {

        // Collection of Character references
        ArrayList<Character> characters = new ArrayList<>();

        // Creating objects
        Warrior w1 = new Warrior("Thor", 150, 40, 80);
        Mage m1 = new Mage("Merlin", 100, 60, 120);

        // Adding objects to collection
        characters.add(w1);
        characters.add(m1);

        // Demonstrating Polymorphism
        for (Character ch : characters) {
            ch.attack();
        }
    }
}

```

Hospital Management System

```
import java.util.ArrayList;

// Base Class
class Patient {
    String name;
    int patientId;
    int age;

    // Constructor
    Patient(String name, int patientId, int age) {
        this.name = name;
        this.patientId = patientId;
        this.age = age;
    }

    // Method
    void displayDetails() {
        System.out.println("Patient Name: " + name);
        System.out.println("Patient ID: " + patientId);
        System.out.println("Age: " + age);
    }
}

// Derived Class - InPatient
class InPatient extends Patient {
    int roomNumber;

    // Constructor
    InPatient(String name, int patientId,
               int age, int roomNumber) {
        super(name, patientId, age);
        this.roomNumber = roomNumber;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- InPatient Details ---");
        System.out.println("Patient Name: " + name);
        System.out.println("Patient ID: " + patientId);
        System.out.println("Age: " + age);
        System.out.println("Room Number: " + roomNumber);
    }
}

// Derived Class - OutPatient
class OutPatient extends Patient {
    String appointmentDate;

    // Constructor
    OutPatient(String name, int patientId,
               int age, String appointmentDate) {
        super(name, patientId, age);
        this.appointmentDate = appointmentDate;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- OutPatient Details ---");
        System.out.println("Patient Name: " + name);
        System.out.println("Patient ID: " + patientId);
        System.out.println("Age: " + age);
        System.out.println("Appointment Date: " + appointmentDate);
    }
}

// Main Class
public class HospitalManagementSystem {
    public static void main(String[] args) {

        // Collection of Patient references
        ArrayList<Patient> patients = new ArrayList<>();

        // Creating objects
        InPatient p1 = new InPatient(
            "Aniket",
            101,
            20,
            305
        );

        OutPatient p2 = new OutPatient(
            "Rahul",
            102,
            22,
            "15-05-2026"
        );

        // Adding objects to collection
        patients.add(p1);
        patients.add(p2);

        // Demonstrating Polymorphism
        for (Patient p : patients) {
            p.displayDetails();
        }
    }
}
```

Hotel Booking System

```
import java.util.ArrayList;

// Base Class
class Room {
    int roomNumber;
    String roomType;
    double pricePerNight;

    // Constructor
    Room(int roomNumber, String roomType, double pricePerNight) {
        this.roomNumber = roomNumber;
        this.roomType = roomType;
        this.pricePerNight = pricePerNight;
    }

    // Method
    void displayDetails() {
        System.out.println("Room Number: " + roomNumber);
        System.out.println("Room Type: " + roomType);
        System.out.println("Price Per Night: " + pricePerNight);
    }
}

// Derived Class - DeluxeRoom
class DeluxeRoom extends Room {
    boolean hasSeaView;

    // Constructor
    DeluxeRoom(int roomNumber, String roomType,
               double pricePerNight, boolean hasSeaView) {
        super(roomNumber, roomType, pricePerNight);
        this.hasSeaView = hasSeaView;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Deluxe Room Details ---");
        System.out.println("Room Number: " + roomNumber);
        System.out.println("Room Type: " + roomType);
        System.out.println("Price Per Night: " + pricePerNight);
        System.out.println("Sea View Available: " + hasSeaView);
    }
}

// Derived Class - SuiteRoom
class SuiteRoom extends Room {
    int numberOfRooms;

    // Constructor
    SuiteRoom(int roomNumber, String roomType,
               double pricePerNight, int numberOfRooms) {
        super(roomNumber, roomType, pricePerNight);
        this.numberOfRooms = numberOfRooms;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Suite Room Details ---");
        System.out.println("Room Number: " + roomNumber);
        System.out.println("Room Type: " + roomType);
        System.out.println("Price Per Night: " + pricePerNight);
        System.out.println("Number of Rooms in Suite: "
                           + numberOfRooms);
    }
}

// Main Class
public class HotelBookingSystem {
    public static void main(String[] args) {

        // Collection of Room references
        ArrayList<Room> rooms = new ArrayList<>();

        // Creating objects
        DeluxeRoom d1 = new DeluxeRoom(
            101,
            "Deluxe",
            4500,
            true
        );

        SuiteRoom s1 = new SuiteRoom(
            202,
            "Suite",
            8500,
            3
        );

        // Adding objects to collection
        rooms.add(d1);
        rooms.add(s1);

        // Demonstrating Polymorphism
        for (Room r : rooms) {
            r.displayDetails();
        }
    }
}
```

Job Recruitment System

```
import java.util.ArrayList;

// Base Class
class JobApplicant {
    String name;
    int applicantId;
    int experienceYears;

    // Constructor
    JobApplicant(String name, int applicantId, int experienceYears) {
        this.name = name;
        this.applicantId = applicantId;
        this.experienceYears = experienceYears;
    }

    // Method
    void displayDetails() {
        System.out.println("Applicant Name: " + name);
        System.out.println("Applicant ID: " + applicantId);
        System.out.println("Experience: " + experienceYears + " yea
    }
}

// Derived Class - FresherApplicant
class FresherApplicant extends JobApplicant {
    double cgpa;

    // Constructor
    FresherApplicant(String name, int applicantId,
                     int experienceYears, double cgpa) {
        super(name, applicantId, experienceYears);
        this.cgpa = cgpa;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Fresher Applicant Details ---");
        System.out.println("Applicant Name: " + name);
        System.out.println("Applicant ID: " + applicantId);
        System.out.println("Experience: " + experienceYears + " yea
        System.out.println("CGPA: " + cgpa);
    }
}

// Derived Class - ExperiencedApplicant
class ExperiencedApplicant extends JobApplicant {
    String previousCompany;

    // Constructor
    ExperiencedApplicant(String name, int applicantId,
                         int experienceYears,
                         String previousCompany) {
        super(name, applicantId, experienceYears);
        this.previousCompany = previousCompany;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Experienced Applicant Details ---
        System.out.println("Applicant Name: " + name);
        System.out.println("Applicant ID: " + applicantId);
        System.out.println("Experience: " + experienceYears + " yea
        System.out.println("Previous Company: " + previousCompany);
    }
}

// Main Class
public class JobRecruitmentSystem {
    public static void main(String[] args) {

        // Collection of JobApplicant references
        ArrayList<JobApplicant> applicants = new ArrayList<>();

        // Creating objects
        FresherApplicant f1 = new FresherApplicant(
            "Aniket",
            101,
            0,
            8.7
        );

        ExperiencedApplicant e1 = new ExperiencedApplicant(
            "Rahul",
            202,
            4,
            "Infosys"
        );

        // Adding objects to collection
        applicants.add(f1);
        applicants.add(e1);

        // Demonstrating Polymorphism
        for (JobApplicant j : applicants) {
            j.displayDetails();
        }
    }
}
```

Library Management System

```
import java.util.ArrayList;

// Base Class
class Book {
    String title;
    String author;
    String ISBN;

    // Constructor
    Book(String title, String author, String ISBN) {
        this.title = title;
        this.author = author;
        this.ISBN = ISBN;
    }

    // Method
    void displayDetails() {
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("ISBN: " + ISBN);
    }
}

// Derived Class - Ebook
class Ebook extends Book {
    double fileSize;

    // Constructor
    Ebook(String title, String author, String ISBN, double fileSize) {
        super(title, author, ISBN);
        this.fileSize = fileSize;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Ebook Details ---");
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("ISBN: " + ISBN);
        System.out.println("File Size: " + fileSize + " MB");
    }
}

// Derived Class - PrintedBook
class PrintedBook extends Book {
    int pageCount;

    // Constructor
    PrintedBook(String title, String author, String ISBN, int pageCount) {
        super(title, author, ISBN);
        this.pageCount = pageCount;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Printed Book Details ---");
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("ISBN: " + ISBN);
        System.out.println("Page Count: " + pageCount);
    }
}

// Main Class
public class LibraryManagementSystem {
    public static void main(String[] args) {

        // Collection of Book references
        ArrayList<Book> books = new ArrayList<>();

        // Creating objects
        Ebook e1 = new Ebook(
            "Java Programming",
            "James Gosling",
            "ISBN101",
            15.5
        );

        PrintedBook p1 = new PrintedBook(
            "Data Structures",
            "Mark Allen",
            "ISBN202",
            450
        );

        // Adding objects to collection
        books.add(e1);
        books.add(p1);

        // Demonstrating Polymorphism
        for (Book b : books) {
            b.displayDetails();
        }
    }
}
```

Music Streaming System

```
import java.util.ArrayList;

// Base Class
class MusicTrack {
    String title;
    String artist;
    double duration;

    // Constructor
    MusicTrack(String title, String artist, double duration) {
        this.title = title;
        this.artist = artist;
        this.duration = duration;
    }

    // Method
    void play() {
        System.out.println("Playing track: " + title);
    }
}

// Derived Class - MP3Track
class MP3Track extends MusicTrack {
    double fileSize;

    // Constructor
    MP3Track(String title, String artist,
              double duration, double fileSize) {
        super(title, artist, duration);
        this.fileSize = fileSize;
    }

    // Overriding Method
    @Override
    void play() {
        System.out.println("\n--- MP3 Track ---");
        System.out.println("Playing MP3 Track: " + title);
        System.out.println("Artist: " + artist);
        System.out.println("Duration: " + duration + " minutes");
        System.out.println("File Size: " + fileSize + " MB");
    }
}

// Derived Class - LiveStreamTrack
class LiveStreamTrack extends MusicTrack {
    String streamQuality;

    // Constructor
    LiveStreamTrack(String title, String artist,
                    double duration, String streamQuality) {
        super(title, artist, duration);
        this.streamQuality = streamQuality;
    }

    // Overriding Method
    @Override
    void play() {
        System.out.println("\n--- Live Stream Track ---");
        System.out.println("Streaming Live Track: " + title);
        System.out.println("Artist: " + artist);
        System.out.println("Duration: " + duration + " minutes");
        System.out.println("Stream Quality: " + streamQuality);
    }
}

// Main Class
public class MusicStreamingSystem {
    public static void main(String[] args) {

        // Collection of MusicTrack references
        ArrayList<MusicTrack> tracks = new ArrayList<>();

        // Creating objects
        MP3Track mp3 = new MP3Track(
            "Shape of You",
            "Ed Sheeran",
            4.2,
            8.5
        );

        LiveStreamTrack live = new LiveStreamTrack(
            "Live Concert",
            "Arijit Singh",
            90,
            "HD"
        );

        // Adding objects to collection
        tracks.add(mp3);
        tracks.add(live);

        // Demonstrating Polymorphism
        for (MusicTrack track : tracks) {
            track.play();
        }
    }
}
```

Online Payment System

```
import java.util.ArrayList;

// Base Class
class Payment {
    String transactionId;
    double amount;
    String date;

    // Constructor
    Payment(String transactionId, double amount, String date) {
        this.transactionId = transactionId;
        this.amount = amount;
        this.date = date;
    }

    // Method
    void processPayment() {
        System.out.println("Processing payment of Rs." + amount);
    }
}

// Derived Class - CreditCardPayment
class CreditCardPayment extends Payment {
    String cardNumber;

    // Constructor
    CreditCardPayment(String transactionId,
                      double amount,
                      String date,
                      String cardNumber) {
        super(transactionId, amount, date);
        this.cardNumber = cardNumber;
    }

    // Overriding Method
    @Override
    void processPayment() {
        System.out.println("\n--- Credit Card Payment ---");
        System.out.println("Transaction ID: " + transactionId);
        System.out.println("Amount: Rs." + amount);
        System.out.println("Date: " + date);
        System.out.println("Card Number: " + cardNumber);
        System.out.println("Payment Processed Successfully!");
    }
}

// Derived Class - UPIPayment
class UPIPayment extends Payment {
    String upId;

    // Constructor
    UPIPayment(String transactionId,
               double amount,
               String date,
               String upId) {
        super(transactionId, amount, date);
        this.upId = upId;
    }

    // Overriding Method
    @Override
    void processPayment() {
        System.out.println("\n--- UPI Payment ---");
        System.out.println("Transaction ID: " + transactionId);
        System.out.println("Amount: Rs." + amount);
        System.out.println("Date: " + date);
        System.out.println("UPI ID: " + upId);
        System.out.println("Payment Processed Successfully!");
    }
}

// Main Class
public class OnlinePaymentSystem {
    public static void main(String[] args) {

        // Collection of Payment references
        ArrayList<Payment> payments = new ArrayList<>();

        // Creating objects
        CreditCardPayment c1 = new CreditCardPayment(
            "TXN101",
            2500,
            "12-05-2026",
            "1234-5678-9876"
        );

        UPIPayment u1 = new UPIPayment(
            "TXN202",
            1500,
            "12-05-2026",
            "aniket@upi"
        );

        // Adding objects to collection
        payments.add(c1);
        payments.add(u1);

        // Demonstrating Polymorphism
        for (Payment p : payments) {
            p.processPayment();
        }
    }
}
```

Online Shopping System

```
import java.util.ArrayList;

// Base Class
class Product {
    int productId;
    String name;
    double price;

    // Constructor
    Product(int productId, String name, double price) {
        this.productId = productId;
        this.name = name;
        this.price = price;
    }

    // Method
    void displayDetails() {
        System.out.println("Product ID: " + productId);
        System.out.println("Name: " + name);
        System.out.println("Price: " + price);
    }
}

// Derived Class - Electronics
class Electronics extends Product {
    int warrantyPeriod;

    // Constructor
    Electronics(int productId, String name,
        double price, int warrantyPeriod) {
        super(productId, name, price);
        this.warrantyPeriod = warrantyPeriod;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Electronics Product Details ---");
        System.out.println("Product ID: " + productId);
        System.out.println("Name: " + name);
        System.out.println("Price: " + price);
        System.out.println("Warranty Period: "
            + warrantyPeriod + " years");
    }
}

// Derived Class - Clothing
class Clothing extends Product {
    String size;
    String color;

    // Constructor
    Clothing(int productId, String name,
        double price, String size, String color) {
        super(productId, name, price);
        this.size = size;
        this.color = color;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Clothing Product Details ---");
        System.out.println("Product ID: " + productId);
        System.out.println("Name: " + name);
        System.out.println("Price: " + price);
        System.out.println("Size: " + size);
        System.out.println("Color: " + color);
    }
}

// Main Class
public class OnlineShoppingSystem {
    public static void main(String[] args) {

        // Collection of Product references
        ArrayList<Product> products = new ArrayList<>();

        // Creating objects
        Electronics e1 = new Electronics(
            101,
            "Laptop",
            65000,
            2
        );

        Clothing c1 = new Clothing(
            201,
            "T-Shirt",
            999,
            "L",
            "Black"
        );

        // Adding objects to collection
        products.add(e1);
        products.add(c1);

        // Demonstrating Polymorphism
        for (Product p : products) {
            p.displayDetails();
        }
    }
}
```

Smart Home Device Management System

```
import java.util.ArrayList;

// Base Class
class SmartDevice {
    String deviceName;
    String brand;
    double powerConsumption;

    // Constructor
    SmartDevice(String deviceName, String brand, double powerConsum
        this.deviceName = deviceName;
        this.brand = brand;
        this.powerConsumption = powerConsumption;
    }

    // Method
    void displayDetails() {
        System.out.println("Device Name: " + deviceName);
        System.out.println("Brand: " + brand);
        System.out.println("Power Consumption: "
            + powerConsumption + " Watts");
    }
}

// Derived Class - SmartLight
class SmartLight extends SmartDevice {
    String brightnessLevel;

    // Constructor
    SmartLight(String deviceName, String brand,
        double powerConsumption, String brightnessLevel) {
        super(deviceName, brand, powerConsumption);
        this.brightnessLevel = brightnessLevel;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Smart Light Details ---");
        System.out.println("Device Name: " + deviceName);
        System.out.println("Brand: " + brand);
        System.out.println("Power Consumption: "
            + powerConsumption + " Watts");
        System.out.println("Brightness Level: " + brightnessLevel);
    }
}

// Derived Class - SmartThermostat
class SmartThermostat extends SmartDevice {
    double temperatureSetting;

    // Constructor
    SmartThermostat(String deviceName, String brand,
        double powerConsumption,
        double temperatureSetting) {
        super(deviceName, brand, powerConsumption);
        this.temperatureSetting = temperatureSetting;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Smart Thermostat Details ---");
        System.out.println("Device Name: " + deviceName);
        System.out.println("Brand: " + brand);
        System.out.println("Power Consumption: "
            + powerConsumption + " Watts");
        System.out.println("Temperature Setting: "
            + temperatureSetting + " C");
    }
}

// Main Class
public class SmartHomeDeviceManagementSystem {
    public static void main(String[] args) {

        // Collection of SmartDevice references
        ArrayList<SmartDevice> devices = new ArrayList<>();

        // Creating objects
        SmartLight light1 = new SmartLight(
            "Smart LED Bulb",
            "Philips",
            12.5,
            "High"
        );

        SmartThermostat thermostat1 = new SmartThermostat(
            "Smart Thermostat",
            "Google Nest",
            25.0,
            24.5
        );

        // Adding objects to collection
        devices.add(light1);
        devices.add(thermostat1);

        // Demonstrating Polymorphism
        for (SmartDevice device : devices) {
            device.displayDetails();
        }
    }
}
```

Student Management System

```
import java.util.ArrayList;

// Base Class
class Student {
    String name;
    int studentId;
    String grade;

    // Constructor
    Student(String name, int studentId, String grade) {
        this.name = name;
        this.studentId = studentId;
        this.grade = grade;
    }

    // Method
    void displayDetails() {
        System.out.println("Student Name: " + name);
        System.out.println("Student ID: " + studentId);
        System.out.println("Grade: " + grade);
    }
}

// Derived Class - UndergraduateStudent
class UndergraduateStudent extends Student {
    String major;

    // Constructor
    UndergraduateStudent(String name, int studentId,
        String grade, String major) {
        super(name, studentId, grade);
        this.major = major;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Undergraduate Student Details ---");
        System.out.println("Student Name: " + name);
        System.out.println("Student ID: " + studentId);
        System.out.println("Grade: " + grade);
        System.out.println("Major: " + major);
    }
}

// Derived Class - GraduateStudent
class GraduateStudent extends Student {
    String researchTopic;

    // Constructor
    GraduateStudent(String name, int studentId,
        String grade, String researchTopic) {
        super(name, studentId, grade);
        this.researchTopic = researchTopic;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Graduate Student Details ---");
        System.out.println("Student Name: " + name);
        System.out.println("Student ID: " + studentId);
        System.out.println("Grade: " + grade);
        System.out.println("Research Topic: " + researchTopic);
    }
}

// Main Class
public class StudentManagementSystem {
    public static void main(String[] args) {

        // Collection of Student references
        ArrayList<Student> students = new ArrayList<>();

        // Creating objects
        UndergraduateStudent u1 = new UndergraduateStudent(
            "Aniket",
            101,
            "A",
            "Computer Engineering"
        );

        GraduateStudent g1 = new GraduateStudent(
            "Rahul",
            201,
            "A+",
            "Artificial Intelligence"
        );

        // Adding objects to collection
        students.add(u1);
        students.add(g1);

        // Demonstrating Polymorphism
        for (Student s : students) {
            s.displayDetails();
        }
    }
}
```

Transport Ticketing System

```
import java.util.ArrayList;

// Base Class
class Ticket {
    int ticketNumber;
    String passengerName;
    double price;

    // Constructor
    Ticket(int ticketNumber, String passengerName, double price) {
        this.ticketNumber = ticketNumber;
        this.passengerName = passengerName;
        this.price = price;
    }

    // Method
    void displayDetails() {
        System.out.println("Ticket Number: " + ticketNumber);
        System.out.println("Passenger Name: " + passengerName);
        System.out.println("Price: " + price);
    }
}

// Derived Class - BusTicket
class BusTicket extends Ticket {
    String busNumber;

    // Constructor
    BusTicket(int ticketNumber, String passengerName,
        double price, String busNumber) {
        super(ticketNumber, passengerName, price);
        this.busNumber = busNumber;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Bus Ticket Details ---");
        System.out.println("Ticket Number: " + ticketNumber);
        System.out.println("Passenger Name: " + passengerName);
        System.out.println("Price: " + price);
        System.out.println("Bus Number: " + busNumber);
    }
}

// Derived Class - TrainTicket
class TrainTicket extends Ticket {
    String coachClass;

    // Constructor
    TrainTicket(int ticketNumber, String passengerName,
        double price, String coachClass) {
        super(ticketNumber, passengerName, price);
        this.coachClass = coachClass;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Train Ticket Details ---");
        System.out.println("Ticket Number: " + ticketNumber);
        System.out.println("Passenger Name: " + passengerName);
        System.out.println("Price: " + price);
        System.out.println("Coach Class: " + coachClass);
    }
}

// Main Class
public class TransportTicketingSystem {
    public static void main(String[] args) {

        // Collection of Ticket references
        ArrayList<Ticket> tickets = new ArrayList<>();

        // Creating objects
        BusTicket b1 = new BusTicket(
            101,
            "Aniket",
            250,
            "MH12AB1234"
        );

        TrainTicket t1 = new TrainTicket(
            202,
            "Rahul",
            850,
            "AC First Class"
        );

        // Adding objects to collection
        tickets.add(b1);
        tickets.add(t1);

        // Demonstrating Polymorphism
        for (Ticket t : tickets) {
            t.displayDetails();
        }
    }
}
```

University Staff Management System

```
import java.util.ArrayList;

// Base Class
class Staff {
    String name;
    int staffId;
    double salary;

    // Constructor
    Staff(String name, int staffId, double salary) {
        this.name = name;
        this.staffId = staffId;
        this.salary = salary;
    }

    // Method
    void displayDetails() {
        System.out.println("Name: " + name);
        System.out.println("Staff ID: " + staffId);
        System.out.println("Salary: " + salary);
    }
}

// Derived Class - Professor
class Professor extends Staff {
    String specialization;

    // Constructor
    Professor(String name, int staffId,
        double salary, String specialization) {
        super(name, staffId, salary);
        this.specialization = specialization;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Professor Details ---");
        System.out.println("Name: " + name);
        System.out.println("Staff ID: " + staffId);
        System.out.println("Salary: " + salary);
        System.out.println("Specialization: " + specialization);
    }
}

// Derived Class - Administrator
class Administrator extends Staff {
    String department;

    // Constructor
    Administrator(String name, int staffId,
        double salary, String department) {
        super(name, staffId, salary);
        this.department = department;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Administrator Details ---");
        System.out.println("Name: " + name);
        System.out.println("Staff ID: " + staffId);
        System.out.println("Salary: " + salary);
        System.out.println("Department: " + department);
    }
}

// Main Class
public class UniversityStaffManagementSystem {
    public static void main(String[] args) {

        // Collection of Staff references
        ArrayList<Staff> staffList = new ArrayList<>();

        // Creating objects
        Professor p1 = new Professor(
            "Dr. Sharma",
            101,
            85000,
            "Computer Science"
        );

        Administrator a1 = new Administrator(
            "Mr. Patil",
            201,
            55000,
            "Examination Department"
        );

        // Adding objects to collection
        staffList.add(p1);
        staffList.add(a1);

        // Demonstrating Polymorphism
        for (Staff s : staffList) {
            s.displayDetails();
        }
    }
}
```

Vehicle Management System

```
import java.util.ArrayList;

// Base Class
class Vehicle {
    String brand;
    String model;
    double price;

    // Constructor
    Vehicle(String brand, String model, double price) {
        this.brand = brand;
        this.model = model;
        this.price = price;
    }

    // Method
    void displayInfo() {
        System.out.println("Brand: " + brand);
        System.out.println("Model: " + model);
        System.out.println("Price: " + price);
    }
}

// Derived Class - Car
class Car extends Vehicle {
    int seatingCapacity;

    // Constructor
    Car(String brand, String model, double price, int seatingCapaci
        super(brand, model, price);
        this.seatingCapacity = seatingCapacity;
    }

    // Overriding Method
    @Override
    void displayInfo() {
        System.out.println("\n--- Car Details ---");
        System.out.println("Brand: " + brand);
        System.out.println("Model: " + model);
        System.out.println("Price: " + price);
        System.out.println("Seating Capacity: " + seatingCapacity);
    }
}

// Derived Class - Bike
class Bike extends Vehicle {
    int engineCC;

    // Constructor
    Bike(String brand, String model, double price, int engineCC) {
        super(brand, model, price);
        this.engineCC = engineCC;
    }

    // Overriding Method
    @Override
    void displayInfo() {
        System.out.println("\n--- Bike Details ---");
        System.out.println("Brand: " + brand);
        System.out.println("Model: " + model);
        System.out.println("Price: " + price);
        System.out.println("Engine Capacity: " + engineCC + " CC");
    }
}

// Main Class
public class VehicleManagementSystem {
    public static void main(String[] args) {

        // Collection of Vehicle references
        ArrayList<Vehicle> vehicles = new ArrayList<>();

        // Creating objects
        Car car1 = new Car("Toyota", "Fortuner", 4500000, 7);
        Bike bike1 = new Bike("Yamaha", "R15", 180000, 155);

        // Adding objects to collection
        vehicles.add(car1);
        vehicles.add(bike1);

        // Demonstrating Polymorphism
        for (Vehicle v : vehicles) {
            v.displayInfo();
        }
    }
}
```

Vehicle Rental System

```
import java.util.ArrayList;

// Base Class
class Vehicle {
    int vehicleId;
    String brand;
    double rentalPricePerDay;

    // Constructor
    Vehicle(int vehicleId, String brand, double rentalPricePerDay) {
        this.vehicleId = vehicleId;
        this.brand = brand;
        this.rentalPricePerDay = rentalPricePerDay;
    }

    // Method
    void displayDetails() {
        System.out.println("Vehicle ID: " + vehicleId);
        System.out.println("Brand: " + brand);
        System.out.println("Rental Price Per Day: Rs." + rentalPricePerDay);
    }
}

// Derived Class - Car
class Car extends Vehicle {
    int numSeats;

    // Constructor
    Car(int vehicleId, String brand,
        double rentalPricePerDay, int numSeats) {
        super(vehicleId, brand, rentalPricePerDay);
        this.numSeats = numSeats;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Car Details ---");
        System.out.println("Vehicle ID: " + vehicleId);
        System.out.println("Brand: " + brand);
        System.out.println("Rental Price Per Day: Rs." + rentalPricePerDay);
        System.out.println("Number of Seats: " + numSeats);
    }
}

// Derived Class - Bike
class Bike extends Vehicle {
    boolean hasHelmet;

    // Constructor
    Bike(int vehicleId, String brand,
        double rentalPricePerDay, boolean hasHelmet) {
        super(vehicleId, brand, rentalPricePerDay);
        this.hasHelmet = hasHelmet;
    }

    // Overriding Method
    @Override
    void displayDetails() {
        System.out.println("\n--- Bike Details ---");
        System.out.println("Vehicle ID: " + vehicleId);
        System.out.println("Brand: " + brand);
        System.out.println("Rental Price Per Day: Rs." + rentalPricePerDay);
        System.out.println("Helmet Available: " + hasHelmet);
    }
}

// Main Class
public class VehicleRentalSystem {
    public static void main(String[] args) {

        // Collection of Vehicle references
        ArrayList<Vehicle> vehicles = new ArrayList<>();

        // Creating objects
        Car c1 = new Car(
            101,
            "Toyota",
            2500,
            5
        );

        Bike b1 = new Bike(
            202,
            "Honda",
            800,
            true
        );

        // Adding objects to collection
        vehicles.add(c1);
        vehicles.add(b1);

        // Demonstrating Polymorphism
        for (Vehicle v : vehicles) {
            v.displayDetails();
        }
    }
}
```